



Construction and application of knowledge graph for urban agglomeration emergency rescue collaboration in earthquake disasters

Lin Zhang^a, Qichen Wang^a, Yanjun Guo^b, Xiangliang Tian^c, Lin Qi^{a,*}

^a College of Management Sciences and Engineering, Beijing Information Science and Technology University, Beijing 100192, China

^b School of Intelligent Engineering and Automation, Beijing University of Posts and Telecommunications, Beijing 100876, China

^c Key Laboratory of Non-coal mine safety risk monitoring and early warning National Mine Safety Administration, China Academy of Safety Science and Technology, Beijing 100012, China

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ABSTRACT

This study proposes an emergency rescue collaboration knowledge graph construction method for urban agglomeration in earthquake disasters. Based on the collection of 22 earthquake disaster emergency plans published on the official websites of multiple cities in the Chengdu-Chongqing urban agglomeration in China, earthquake disaster emergency rescue data from the Red Cross Society of Sichuan Province and Chongqing City, and historical rescue information from the China Blue Sky rescue team, this study defines and extracts six types of entities including rescue entities, policy documents, rescue actions, rescue information, rescue supplies, and emergency response levels. A knowledge graph pattern layer is established using a hybrid approach of top-down and bottom-up, including concept layer, relationship layer, rule layer, and instance layer. This study extracts earthquake disaster emergency rescue collaboration knowledge information from collected data sources, and YEDDA software is used for knowledge fusion, thus constructing a knowledge graph data layer. The data is stored in the Neo4j graph database as triplets (entity-relation-entity). Visual representation and retrieval are used to achieve the query, association, and inference of emergency rescue collaboration information for urban agglomeration in earthquake disasters. The 2022 Luding earthquake disaster in Ganzi Tibetan Autonomous Prefecture, China is selected as a typical case, and verified the effectiveness and reliability by inputting the case into the emergency rescue collaboration knowledge graph which was constructed in this study. The results indicate that the constructed knowledge graph provides intelligent decision support for earthquake disaster emergency rescue collaboration in urban agglomeration, effectively improves the performance of earthquake disaster emergency rescue, and provides new ideas and methods for earthquake disaster rescue and reduction.

1. Introduction

China is one of the countries most frequently and severely affected by earthquake disasters in the world. These disasters often exhibit characteristics such as high complexity, significant destructiveness, wide coverage, and unpredictability, which have severe consequences for urban operations and human activities [1]. The golden rescue time for earthquake disasters is within 72 h after the event. Therefore, both emergency command centers and frontline rescue personnel need to make critical decisions in a short period and execute precise emergency rescues swiftly to save more lives and minimize property damage. Additionally, in recent years, China's urbanization development has

gradually shifted from a decentralized development model of small and medium-sized cities to an innovative urban agglomeration development model centered around core cities [2]. While this approach accelerates urbanization, it also reduces the "buffer space" between cities, increasing the complexity of emergency management in urban agglomeration. Thus, in the face of major natural disasters such as earthquakes, enhancing government emergency management capabilities and building more disaster-resilient modern urban agglomeration have become urgent and critical challenges.

Due to the increasing coupling of earthquake disasters, effective collaboration among emergency rescue participants is crucial for successful emergency management [3]. This necessitates efficient

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* Corresponding author.

E-mail addresses: zhanglin@bistu.edu.cn (L. Zhang), 2023020816@bistu.edu.cn (Q. Wang), gjy811@bupt.edu.cn (Y. Guo), xiangliangtian@foxmail.com (X. Tian), qilin@bistu.edu.cn (L. Qi).

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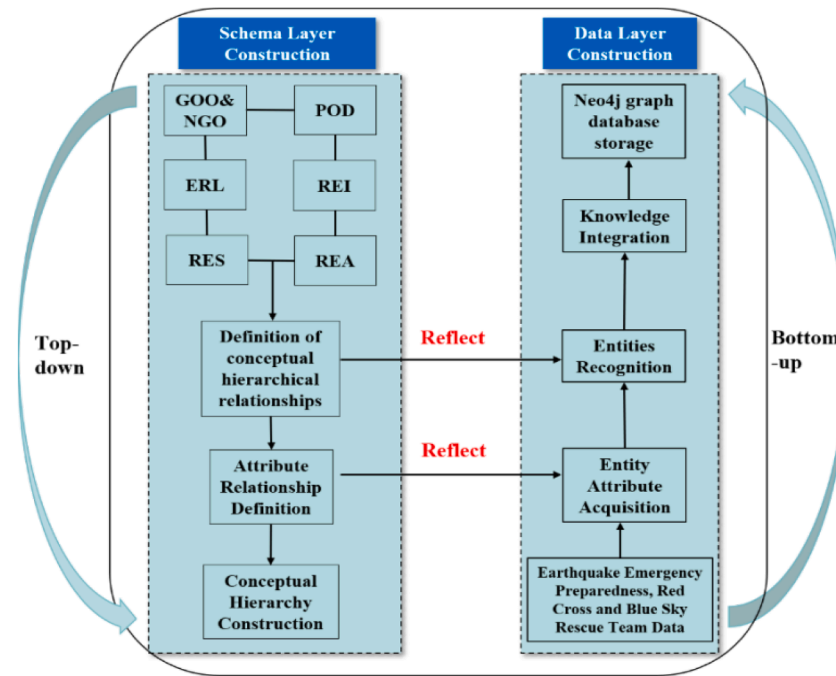


Fig. 1. Flowchart of construction for the earthquake disaster emergency rescue collaboration knowledge graph.

communication across the entire urban agglomeration [4]. Therefore, intelligent knowledge management plays an essential role in enhancing the performance of emergency rescue collaboration [5]. Knowledge graphs describe entities, concepts, and their relationships in the real world. A knowledge graph systematically constructs real-world events into factual representations, capturing the innerrelationships between various information and knowledge elements [6,7]. In recent years, knowledge graph technology has been widely used and disseminated in several fields such as finance, medicine, technology, and production scheduling [8], and has also been explored and researched within the domain of emergency management [9]. For instance, Chen et al. utilize textual data from natural gas pipeline station accidents under harsh environmental conditions to propose a station accident evolution warning model based on knowledge graphs, which enables early warning and accident path prediction [10]. In order to enhance the accuracy of emergency management execution in long-distance oil and gas pipeline accidents, Chen et al. [11] establish an emergency task recommendation model adaptable to multilevel accident information descriptions by aggregating adjacent nodes and relationships in the knowledge graph for embedding learning. Yan et al. [12] develop a subsequent multievent graph convolutional network (SMEGCN) using knowledge graph technology for a series of chain reactions triggered by public health emergencies, which provides practical suggestions for emergency management of public health emergencies. Li et al. [13] construct a multimodal knowledge graph about urban flood emergency knowledge management model to enhance the decision-making capabilities of emergency management departments and improve public emergency evacuation efficiency, thereby reducing losses caused by urban flood disasters. Francesco et al. [14] construct a knowledge graph to provide a comprehensive safety perspective for hazard data, thereby enabling the assessment of report completeness. Wang et al. [15] propose a modeling method for correlation analysis of railway accident hazards based on knowledge graph theory, which describes the relationships between accidents and potential risks within the knowledge graph network to reveal underlying patterns of accident occurrence. Yang et al. [16] develop the COVID-19 public health evidence knowledge graph (CPHEKG). The evidence mapping in CPHEKG can demonstrate the effects of different non-pharmaceutical interventions on the

COVID-19 pandemic. The analysis verifies the quality of our knowledge graph and indicates its potential to support public health policy for COVID-19. Tao et al. [17] apply a standard digital knowledge graph to emergency decision making. The overarching objective is to provide emergency decision makers with an effective support mechanism that delivers both comprehensive information and situation specific solutions tailored to diverse scenarios.

Although previous research has played a significant role in improving urban emergency rescue performance, there remains a gap in studies on applying knowledge graph technology to emergency rescue collaboration for earthquake disasters in urban agglomeration. On one hand, some studies focus on the application of knowledge graph technology within a single city or regional urban emergency management, rarely considering the severe impacts of large-scale natural disasters or major emergencies on the entire urban agglomeration [18–20]. On the other hand, certain studies have applied knowledge graph technology to investigate only a single aspect of emergency rescue. In reality, the key factors affecting emergency rescue during earthquake disasters are often complex and diverse. This situation typically leads to a lack of information transfer between knowledge models, thereby hindering the formulation of emergency rescue decisions [21,22].

Based on this, the study comprehensively considers the difficulties and challenges faced by urban agglomeration in emergency rescue collaboration under current earthquake disasters. It compiles 22 Earthquake Emergency Plans published on the official websites of cities within the Chengdu-Chongqing urban agglomeration, along with earthquake-related data and materials released by the Red Cross Society and the Blue Sky rescue team of Sichuan Province and Chongqing City. Utilizing the Named Entity Recognition method in natural language processing, the study extracts entity information related to urban agglomeration earthquake disaster emergency rescue collaboration and constructs a knowledge graph based on the relationships between entities. The structured data is stored in the Neo4j graph database in the form of triples (entity-relationship-entity), enabling both visual representation and information query retrieval. Subsequently, by inputting the 2022 Luding earthquake disaster event in Ganzi Tibetan Autonomous Prefecture, the effectiveness of the constructed knowledge graph model is validated. This study expands the application of knowledge

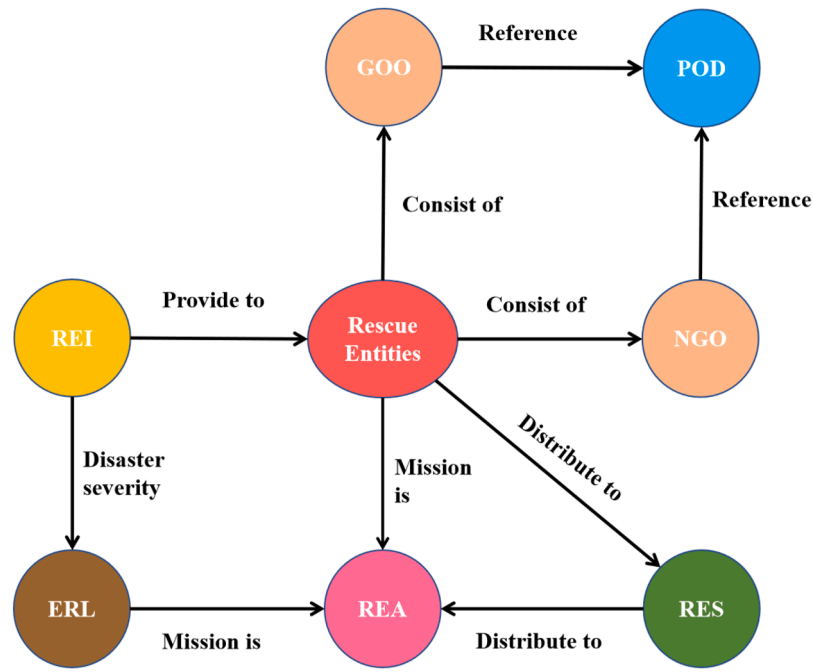


Fig 2. Schematic diagram of entity structure.

graph technology to urban agglomeration earthquake disaster emergency rescue, providing support for making timely and effective decision-making for collaborative emergency response, to enhance urban agglomeration emergency rescue performance and strengthen urban agglomeration safety resilience.

The study is structured as follows. Sections 2 and 3 outline the schema layer and data layer of the urban agglomeration earthquake disaster emergency rescue collaboration knowledge graph, respectively. Section 4 addresses the visualization of the knowledge graph. In Section 5, this study applies the constructed model to a typical case of the Luding earthquake disaster which happened in China. Section 6 concludes the study with key insights, along with a discussion of its limitations and future research directions.

2. Construction of the schema layer for knowledge graph

2.1. Method and process for knowledge graph construction

In the process of constructing a knowledge graph, methods such as bottom-up, top-down, or a combination of both are typically employed. The hybrid construction method, which integrates both approaches, involves predefining the schema layer at the initial stage. The schema layer is upgraded under the guidance of the knowledge organization structure of the data source, resulting in a knowledge organization structure with higher confidence, and making the knowledge graph more complete and more reliable [23]. Considering the advantages and limitations of these three methods, this study adopts a hybrid approach to construct the urban agglomeration earthquake disaster emergency rescue collaboration knowledge graph. The specific process is illustrated in Fig. 1.

The construction process includes knowledge extraction, knowledge fusion, and knowledge structure design. Based on these steps, triple modeling, knowledge storage, and knowledge graph visualization and retrieval are implemented to construct the earthquake disaster emergency rescue collaboration knowledge graph. The materials from the Earthquake Emergency Plans, the Red Cross Society, and the Blue Sky

rescue team selected in this study are unstructured data, requiring knowledge extraction before knowledge fusion can be performed.

After completing the earthquake disaster knowledge extraction, knowledge fusion is performed to eliminate redundancy, thereby refining the definitions of entities to be more comprehensive and precise. Subsequently, the knowledge structure is designed, including schema layer construction (concept layer, relationship layer, rule layer, and instance layer) and data layer construction. Finally, triple modeling is completed, dividing earthquake disaster knowledge into entity-relationship-entity, allowing the knowledge to be stored in Neo4j in the form of triples to complete the earthquake disaster emergency rescue collaboration knowledge graph, and enabling its visualization and information retrieval.

2.2. Ontology design of the knowledge graph

Given the multiplicity of rescue entities and the complexity of operational tasks following earthquake disasters, this study defines the entities within the knowledge graph into six specific categories: rescue entities (Government Organizations and Non-Government Organizations), policy documents, rescue supplies, rescue actions, emergency response levels, and rescue information. The selection of these six categories as entities enables the government to make more informed and effective decisions. It facilitates coordination of the overall situation based on policy documents, directs appropriate rescue actions according to the emergency response level, and supports the execution of rescue operations by relevant entities, while providing vital rescue information and supplies. This approach enhances the government's ability to make quicker and more accurate decisions, rapidly assess the disaster situation, deliver necessary materials to affected populations, reduce casualties and economic losses, and improve overall rescue efficiency.

Since the quadruple representation method, which encompasses concepts, relationships, rules, and instances [24], provides a comprehensive description of earthquake disasters and their emergency rescue collaboration responses, this study selects the quadruple as the descriptive framework for entities. The Earthquake Disaster Emergency

Table 1
Labeling system label description.

Label	Definition	Label	Definition
B-GOO	Starting position of the GOO	B-NGO	Starting position of the NGO
I-GOO	Intermediate or end positions of the GOO	I-NGO	Intermediate or end positions of the NGO
B-POD	Starting position of the POD	B-ERL	Starting position of the ERL
I-POD	Intermediate or end positions of the POD	I-ERL	Intermediate or end positions of the ERL
B-RES	Starting position of the RES	B-REI	Starting position of the REI
I-RES	Intermediate or end positions of the RES	I-REI	Intermediate or end positions of the REI
B-REA	Starting position of the REA	O	Ignored characters
I-REA	Intermediate or end positions of the REA		

Rescue Ontology (EDERO) is defined as the following quadruple, as presented in Eq. (1).

level response, which corresponds to different rescue actions. Categorizing rescue actions in chronological order enables rescue entities to

$$EDERO = \langle EDERO - Concepts, EDERO - Relations, EDERO - Rules, EDERO - Instances \rangle \quad (1)$$

where *EDERO-Concepts* represents the collection of concepts of entities, *EDERO-Relations* represents the collection of conceptual relationships of entities, *EDERO-Rules* represents the collection of rules that objectively exist for entities, and *EDERO-Instances* represents the collection of instances of entities.

Fig. 2 presents a schematic diagram of the entity structure, which includes the entity types and relationship types defined in this study.

The YEDDA software performs text annotation through command lines and keyboard shortcuts, offering intelligent recommendations by learning from the latest annotated texts, thereby overcoming the inefficiency of traditional text annotation methods [25]. Therefore, this study utilizes the YEDDA annotation software to annotate text related to earthquake disaster emergency rescue collaboration knowledge. The annotation focuses on government organizations (GOO) and non-governmental organizations (NGO) within rescue entities, as well as policy documents (POD), rescue supplies (RES), rescue information (REI), rescue actions (REA), and emergency response levels (ERL). The definition of non-governmental organizations in this article is as follows: “Non-governmental organizations are third-party organizations that are not part of the government or the corporate sector. They typically represent social forces and provide public services [26].” Text annotation usually utilizes the beginning-inside-outside (BIO) sequence labeling method, which is encoded as B-(beginning of a noun phrase), I-(inside a noun phrase), and O (outside a noun phrase) [27]. The detailed labeling scheme for text annotation in this study is presented in Table 1.

2.2.1. Concept layer

The core concepts of earthquake disasters are derived from 22 Earthquake Emergency Plans issued by the Chengdu-Chongqing urban agglomeration, as well as relevant materials from the Red Cross Society and the Blue Sky rescue team. These concepts are further expanded accordingly. The concept layer includes six categories of entities, including rescue entities, policy documents, rescue actions, rescue supplies, emergency response levels, and rescue information, as presented in Table 2.

In order to modify the opinions of the evaluation experts, the rescue actions will be refined and categorized according to the chronological order of rescue, initially, according to the earthquake emergency plan “disaster report – prior disposal – organization, command and emergency response – emergency response – end of emergency response – restoration and reconstruction” time sequence of the rescue actions for the initial classification. The second classification of “organization, command and emergency response” is carried out according to the emergency response level, from the first level response to the fourth

more effectively execute their tasks, while clarifying the workflow at different stages of the operation enhances overall rescue efficiency.

2.2.2. Relationship layer

The input of the knowledge graph is primarily in the form of triples, which requires that the relationships between two elements be clearly defined to construct the corresponding semantic network. By analyzing the relationships between entities within materials related to Earthquake Emergency Plans, the Red Cross Society, and the Blue Sky rescue team, and performing clustering operations, the specific concepts of entities and the descriptions of relationships are summarized, as shown in Table 3.

2.2.3. Rule layer

Rules are used as the basis for reasoning at a later stage, and the rule layer is used to define the constraints between concepts and concepts, concepts and relationships. In this study, the rule layer is based on policy documents such as the National Earthquake Emergency Response Plan and the Outline of Chengdu-Chongqing Urban Agglomeration Twin Cities Economic Circle Construction Plan. The earthquake magnitude is entered after an earthquake disaster occurs, and the emergency response level is categorized into first level response, second level response, third level response, and fourth level response according to the high and low magnitude of the earthquake, as well as the casualties caused. First level response requires other cities to carry out coordinated rescue and provide support such as rescue teams, medical and epidemiological prevention, and rescue supplies due to the high level of the earthquake and the large loss of people and property caused by the earthquake. For example, in the 2008 Wenchuan earthquake of magnitude 8, the Chengdu Military Region initiated an emergency response eight minutes after the earthquake, ordering the reserve militia in the disaster area to immediately carry out relief work, as well as dispatching medical emergency units to rescue the injured. Chengdu City Fire Brigade was involved in the rescue of Juyuan Middle School, rescued 55 survivors, dug out more than 300 victims, medical support group first from Sichuan Province and other medical institutions to draw 521 medical technical backbone and the formation of three first aid stations. The special and critically injured were concentrated in three large general hospitals, Sichuan Provincial People’s Hospital, West China Hospital of Sichuan University, and Chengdu General Hospital of Chengdu Military Region, for systematic treatment. A total of 180,000 disaster relief tents were deployed [28]. If located in a densely populated area or after the occurrence of seismic disasters, such as other geologic hazards, the need to start the regional earthquake disaster preparedness joint prevention and control mechanism, other surrounding cities need to carry out

Table 2
Entity concept collection.

First level concepts	Second level concepts	Third level concepts
Rescue entities	GOO	Ministry of Emergency Management of the People's Republic of China, Emergency Management Department, Health Commission, Municipal Earthquake Relief Command, etc.
	NGO	Red Cross Society, Blue Sky rescue team, etc.
Policy documents	Country level	"Emergency Response Law of the People's Republic of China", "National Earthquake Emergency Plan", "Opinions on Establishing and Improving the Regional Collaboration Mechanism for the Red Cross Society Emergency Rescue System" and "Charter of the Red Cross Society of China"
	City level	"Outline of the Construction Plan for the Chengdu-Chongqing Urban Agglomeration Economic Circle", etc.
Rescue supplies	Basic necessities	Daily necessities, food, drinking water, folding beds, etc.
	Medical supplies	Medicines, first aid kits, medical equipment, disinfectants, etc.
Rescue actions	Red Cross Society supplies	Disaster relief family boxes, money
	Disaster report	Collecting, analyzing and judging disaster situations and reporting them in a timely manner
	Prior disposal	Understand the disaster situation, carry out disaster assessment, mobilize rescue teams, and gather support forces
	Organization, command and emergency response	Activate the corresponding emergency response level according to the magnitude of the earthquake, and carry out earthquake relief work
	Emergency response	Earthquake information collection, analysis, and judgment, personnel search and rescue, medical treatment, intensity assessment and damage assessment, etc
	End of emergency response	Termination of emergency rescue operations
Emergency response levels	Restoration and reconstruction	Rehabilitation and reconstruction, mass resettlement
	By the degree of disaster damage	First level response, second level response, third level response, fourth level response
Rescue information	Earthquake information	Provided by the Earthquake Agency/Agency or the Institute of Earthquake Forecasting
	Disaster assessment information	Provided by the Emergency Management Bureau, Earthquake Agency
	Rescue team and resource information	Provided by the Emergency Management Bureau
	Communications and transportation information	Provided by the Transportation Bureau

emergency coordinated rescue work. If the disaster is in a largely controllable situation, the same actions as the fourth level response, the urban agglomeration within the autonomous rescue, material dispatch, medical assistance and other work. The construction of the rule layer is illustrated in Fig. 3.

2.2.4. Instance layer

Instances represent the concrete expressions of ontological concepts. In this study, the instance layer is demonstrated using rescue entities as an example. At the municipal level, rescue entities are divided into

Table 3
Entity relationship aggregation.

Relationship	Relationship description
Governance	Governance mainly refers to a kind of administrative division affiliation. For example, a province under the jurisdiction of a number of prefecture level cities (states), Sichuan Province under the jurisdiction of the city of Chengdu, Mianyang City, etc
Disaster severity	Relationship between rescue information and emergency response levels
Reference	Relationship between rescue entities and policy documents
Classified	Internal organizational structure of the rescue entities
Mission is	Relationship between rescue entities and rescue actions
Provide to	Relationship between rescue information and rescue entities
Distribute to	Relationship between rescue entities, rescue actions and rescue supplies
Command	Relationship between the Province or City and the rescue entities
Instances	Specifics within the rescue entities
Follow up	Time sequence between rescue actions
Consist of	Interior of the rescue entities(divided into NGO and GOO)

governmental organizations and non-governmental organizations (NGO). Governmental organizations include entities such as the Earthquake Agency, the Health Commission, and others, while NGO include the Red Cross Society and the Blue Sky rescue team, as shown in Fig. 4.

The Red Cross Society of Sichuan Province is organized into several collaborative zones, such as the Eastern Sichuan Collaboration Zone, Western Sichuan Collaboration Zone, and Southern Sichuan Collaboration Zone, which can respond to rescue efforts for major natural disasters like earthquakes. These collaboration zones have conducted numerous drills, playing a significant role in coordinated rescue actions within urban agglomerations. The specific internal structure of the Red Cross Society is illustrated in Fig. 5.

3. Construction of the data layer for knowledge graph

The construction process of the data layer mainly includes four components: data collection, knowledge extraction, knowledge fusion, and knowledge application, as illustrated in Fig. 6. In this study, data collection includes 22 Earthquake Emergency Response Plans published by various cities in China's Chengdu-Chongqing urban agglomeration, along with documentation from the Red Cross Society branches in Sichuan Province and Chongqing City, as well as the Blue Sky rescue team, totaling 71,290 Chinese characters. The first step involves annotating the textual data using the BIO tagging method. The second step is knowledge extraction, where the data is transformed into entity-relationship-entity triplets. The third step is knowledge fusion, which unifies various expressions corresponding to the same entity to enhance data accuracy. The fourth step is knowledge application, which involves visualizing and querying the constructed knowledge graph.

3.1. Knowledge extraction

Data sources for seismic hazards are in various forms, and currently the common data types are divided into three categories: structured data, semi-structured data and unstructured data. Unstructured data is characterized by its high authority, clear emergency response and rescue procedures, and coordination of various forces, which helps optimize resource allocation. However, directly extracting entities and attributes from Earthquake Emergency Plans, Red Cross Society materials, and Blue Sky rescue team documents is challenging. It is necessary to first process the textual materials, converting unstructured data into a structured text format and performing word segmentation. Subsequently, entities and attributes are extracted to form entity-relationship-entity triplets.

Named Entity Recognition (NER) is now widely used by academics in the field of natural language processing. Among the various models, the BERT-BiLSTM-CRF framework and the Hidden Markov Model (HMM)

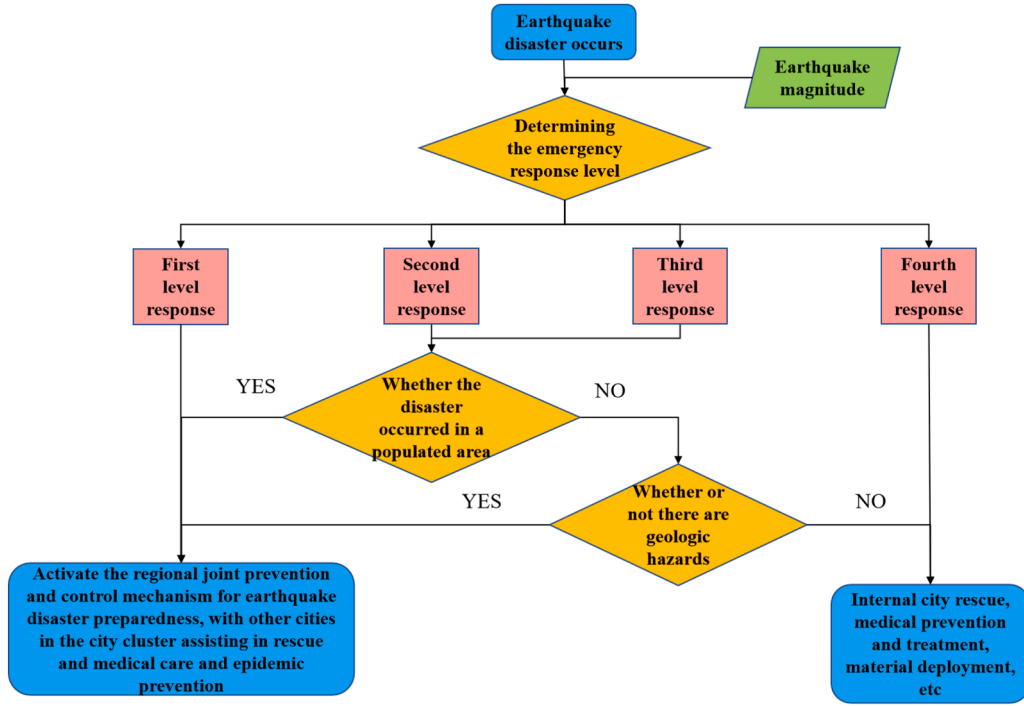


Fig. 3. Rule layer architecture for emergency rescue collaboration.

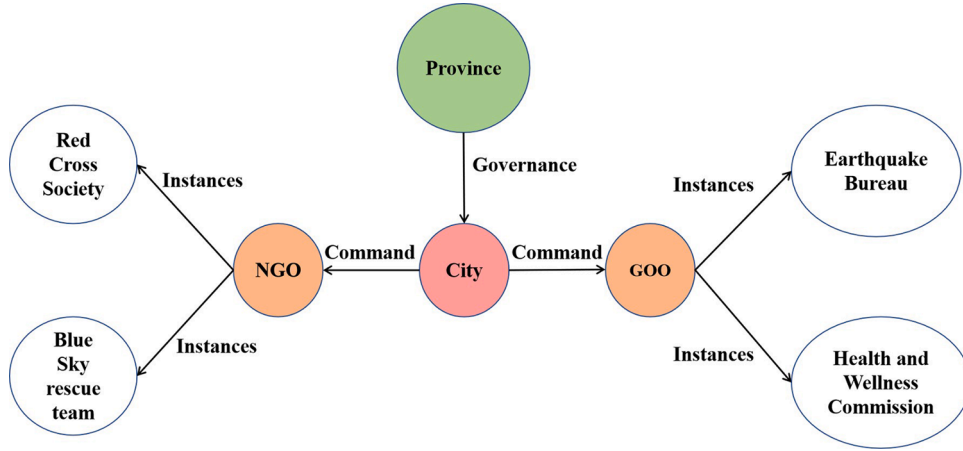


Fig 4. Demonstration of the instance layer.

are more common, and their overall performance on multiple datasets in English and Chinese is better than other frameworks [29]. The Bidirectional Encoder Representations from Transformers (BERT) model can provide a wide range of language coverage, and thus is very favourable for non-English items, and it is better to use the BERT model to run the Chinese text, Bidirectional Long-term and Short-term Memory Network (BiLSTM) model can capture front-to-back and back-to-front related information, and make use of contextual relationships to better obtain bidirectional semantic dependencies, Conditional Random Field (CRF) model can efficiently compute the optimal paths through a dynamic planning algorithm, which can efficiently compute the optimal path and can improve the efficiency of Named Entity Recognition, HMM model can be used to predict the lexical labels of each word given a sequence of words. Therefore, in this study, the above models are utilized for entity recognition on annotated data. The structure of the BERT-BiLSTM-CRF model is schematically shown in Fig. 7.

3.1.1. BERT-BiLSTM-CRF model

The BERT model is designed to capture the contextual relationships between words (or subwords) in text [30]. Its main architecture consists of transformer encoders [31]. The attention mechanism within the encoder simulates human cognition by dynamically adjusting weight coefficients based on the semantic relevance between words in a sentence, thereby generating contextually enriched word representations. This mechanism enables the model to autonomously learn and prioritize salient information in the input [32], as shown in Eq. (2).

$$Attention(Q, K, V) = \text{Softmax}\left(\frac{Q \times K^T}{\sqrt{d_k}}\right)V \quad (2)$$

where Q (Query), K (Key), V (Value) is the word vector matrix, and $\sqrt{d_k}$ is the vector dimension, the purpose of dividing by $\sqrt{d_k}$ is to prevent the vector scores from increasing as the vector dimension increases. So, BERT is able to better recognize semantic relations, more comprehensively describe the polysemy of words, and more accurately infer the

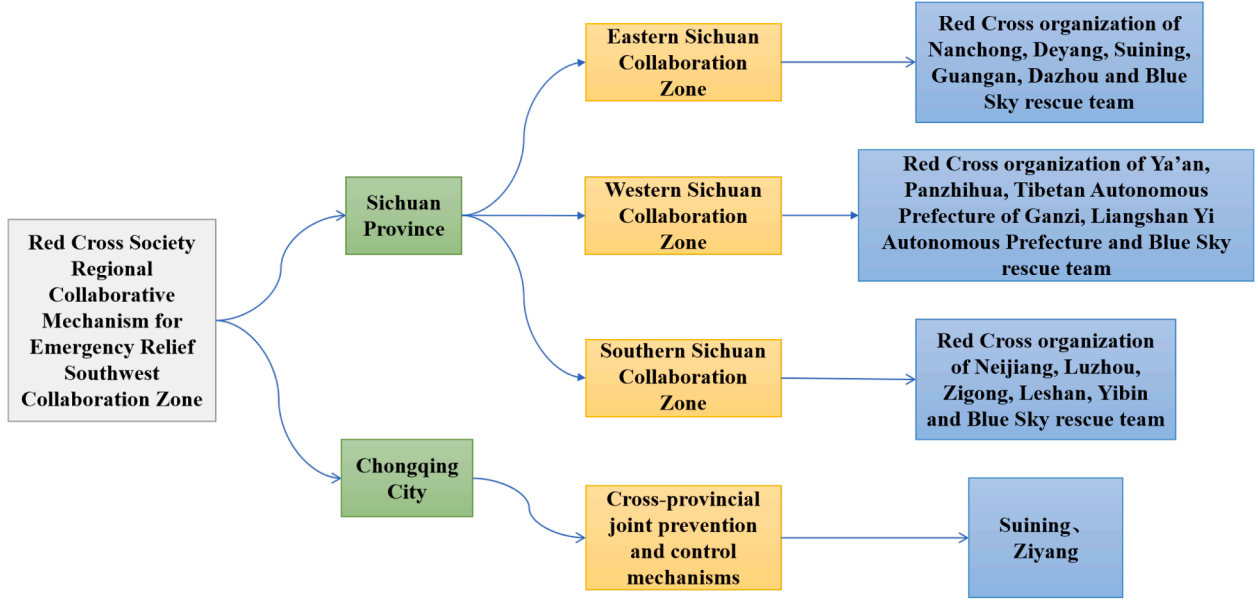


Fig. 5. Example section of the Red Cross Society.

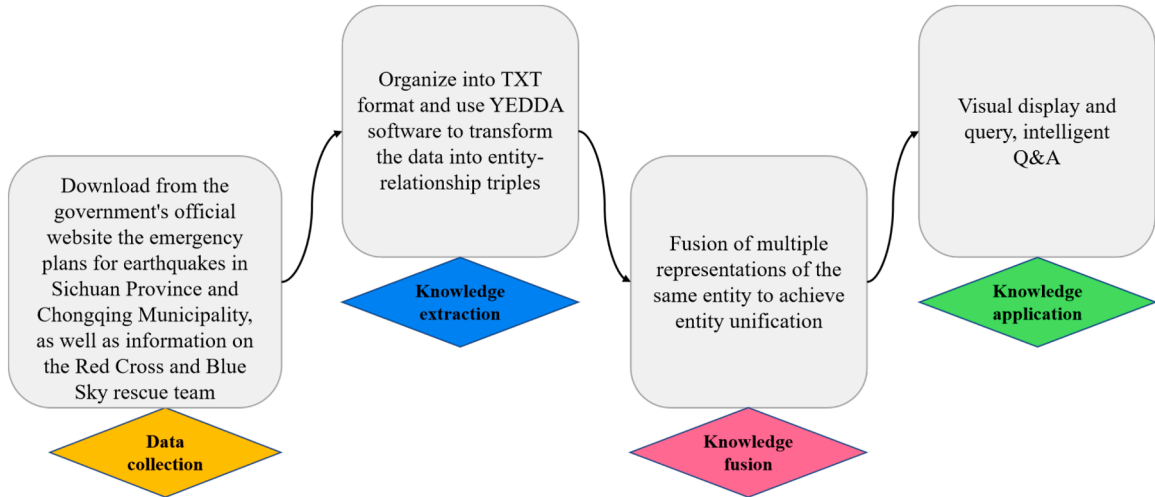


Fig. 6. Construction process of the knowledge graph data layer.

sequences and interrelationships between multiple sentences.

Long-Short Term Memory (LSTM) cleverly utilizes the concept of gating to achieve long-term memory and solves the gradient explosion and gradient vanishing problems during Recurrent Neural Network (RNN) training [33]. Input gates, forgetting gates and output gates are also added for deleting and modifying records, adding records and updating the current short-term memory, respectively, which can trade off the data accordingly [34]. However, LSTM can only capture data in a single direction and cannot obtain comprehensive data and semantic features. The specific computational procedure of LSTM is shown in Eq. (3):

$$\begin{aligned} i_t &= \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \\ f_t &= \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \\ o_t &= \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \\ g_t &= \tanh(W_g \cdot [h_{t-1}, x_t] + b_g) \\ C_t &= f_t \times C_{t-1} + i_t \times g_t \end{aligned} \quad (3)$$

where i denotes the input gate, f denotes the forgetting gate, o denotes the output gate, σ denotes the sigmoid activation function, $\tanh$ denotes the hyperbolic tangent function, W denotes the weight matrix, b denotes

the bias term, h denotes the state of the hidden layer, x_t is the input data at the moment t , g_t is the temporary neuron state, C denotes the neuron state, and the LSTM model schematic is shown in Fig 8.

BiLSTM consists of two parts, which are a forward LSTM and a backward LSTM. The forward LSTM passes information in the forward direction, while the backward LSTM propagates information in the reverse direction. Afterwards, the bidirectional LSTM at the same moment is merged to better capture the semantic dependency of text data in both directions, which will significantly improve the accuracy of text data analysis. The final hidden layer state at moment t is shown in Eq. (4):

$$h_t = \vec{h}_t \oplus \overleftarrow{h}_t \quad (4)$$

Conditional Random Field (CRF) can obtain the optimal sequence of label predictions by capturing the dependencies between labels [35], and the accuracy of the relationships between labels can be ensured by using the CRF model. The CRF model is a special case of Markov Random Fields, in which it is assumed that a position in a random field is only related to the assignments of the positions that are adjacent to it, and is

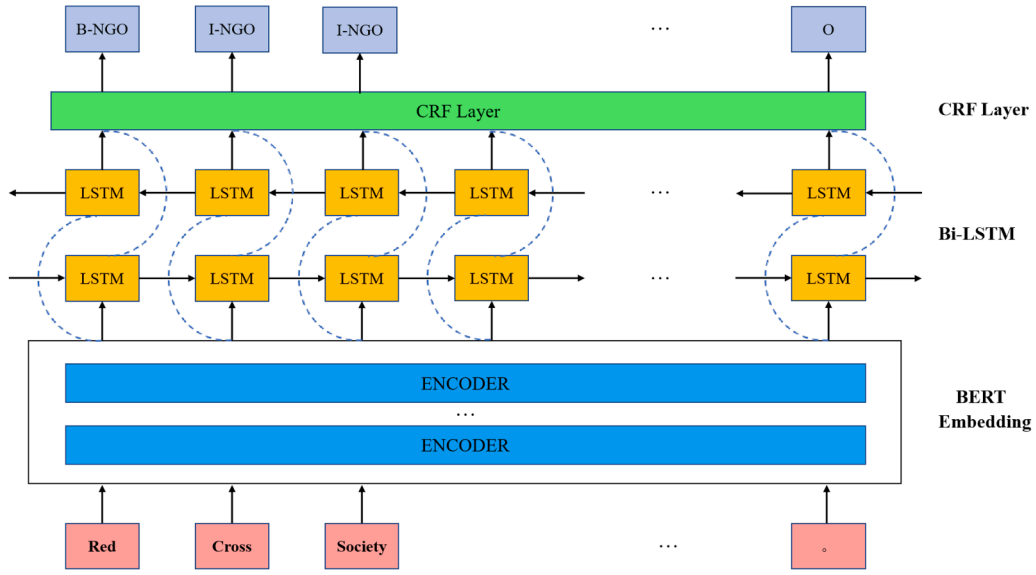


Fig. 7. Structure of the BERT-BiLSTM-CRF model.

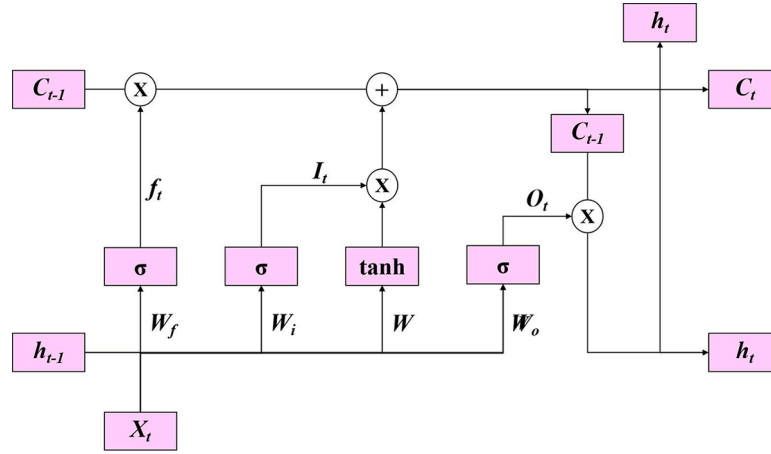


Fig. 8. LSTM model structure.

independent of the assignments of the positions that are not adjacent to it. In the CRF model, it is assumed that the Markov Random Field consists solely of X and Y random variables. For an input sequence $S = \{x_1, x_2, \dots, x_n\}$ and its corresponding sequence of predicted labels $y = \{y_1, y_2, \dots, y_n\}$ where n is the length of the sequence, the CRF model score is shown in Eq. (5):

$$Score(X, y) = \sum_{i=1}^n (W_{y_i, y_{i+1}} + P_{i, y_i}) \quad (5)$$

where W denotes the state transfer matrix, $W_{y_i, y_{i+1}}$ denotes the

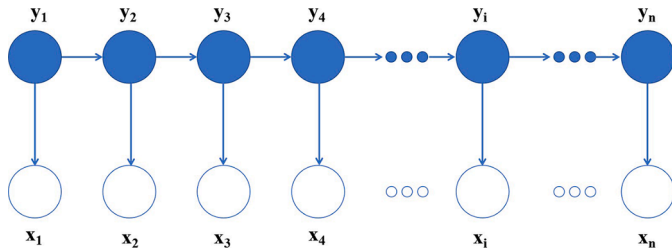


Fig. 9. Schematic of the HMM model.

probability score for a tag y_i to be transferred to a tag y_{i+1} , and P_{i, y_i} denotes the probability score y_i for the i th character to be labeled as a tag. The probability that the input sequence S produces a tagged sequence is shown in Eq. (6), where $\tilde{y}$ denotes a true tag.

$$p(y|S) = \frac{e^{S(X, y)}}{\sum_{\tilde{y} \in Y_x} Score(X, \tilde{y})} \quad (6)$$

The likelihood function for the sequence of training process markers is shown in Eq. (7), where Y_x denotes the set of all markers.

$$\log(p(y|S)) = Score(X, y) - \log \left(\sum_{\tilde{y} \in Y_x} e^{Score(X, \tilde{y})} \right) \quad (7)$$

Finally, the set of output sequences y^* with the largest overall probability score is obtained, and the final sequence is shown in Eq. (8):

$$y^* = \underset{\tilde{y} \in Y_x}{\operatorname{argmax}} Score(X, \tilde{y}) \quad (8)$$

3.1.2. HMM model

Hidden Markov Model (HMM) is a probabilistic model for time series data, which describes a process where an unobservable state sequence is

Table 4
Comparison of experiments.

Model	Precision	Recall	F1
HMM	80.36 %	71.03 %	73.39 %
BERT	60.41 %	85.79 %	70.90 %
BiLSTM-CRF	81.48 %	77.91 %	79.33 %
BERT-BiLSTM-CRF	81.03 %	86.21 %	83.54 %

Table 5
Comparison of knowledge fusion.

Pre-fusion	Post-fusion
End of emergency, end of emergency response	End of emergency response
Medical Epidemiology Team, Medical and Health Epidemiology Team	Medical Epidemiology Team
Level 1 response, first level response, earthquake emergency level 1 response	First level response
...	...

randomly generated by a hidden Markov chain. Each state subsequently emits an observation, thereby producing a sequence of random observations. The state sequence randomly generated by the hidden Markov chain is referred to as the state sequence, denoted as y , each state emits an observation, and the resulting random sequence of observations is called the observation sequence, denoted as x . The structure of the HMM

is illustrated in Fig. 9.

3.1.3. Experimental results

In this study, the comprehensive evaluation index F1 is used as a parametric criterion for the model recognition effect [36]. F1 represents the harmonic mean of Precision (P), which refers to the proportion of correctly labeled data in the results, and Recall (R), which denotes the proportion of all positive examples that are correctly identified as positive [37]. The F1 score is calculated as shown in Eq. (9):

$$P = \frac{\text{Predict number of correct answers}}{\text{The predicted number}}$$

$$R = \frac{\text{The number of correct predictions}}{\text{The number of manual annotations}}$$

$$F1 = \frac{2 \times P \times R}{(P + R)} \quad (9)$$

To evaluate the effectiveness of the BERT-BiLSTM-CRF model in Named Entity Recognition modelling, this study comparative experiments of the HMM model, the BERT model, the BiLSTM-CRF model, and the BERT-BiLSTM-CRF model are carried out in this study. The datasets used for the four models are the same. The experimental results are shown in Table 4.

The experimental comparison reveals that there are some differences in the effectiveness of the four models in Named Entity Recognition. In terms of Precision, the BiLSTM-CRF and BERT-BiLSTM-CRF models

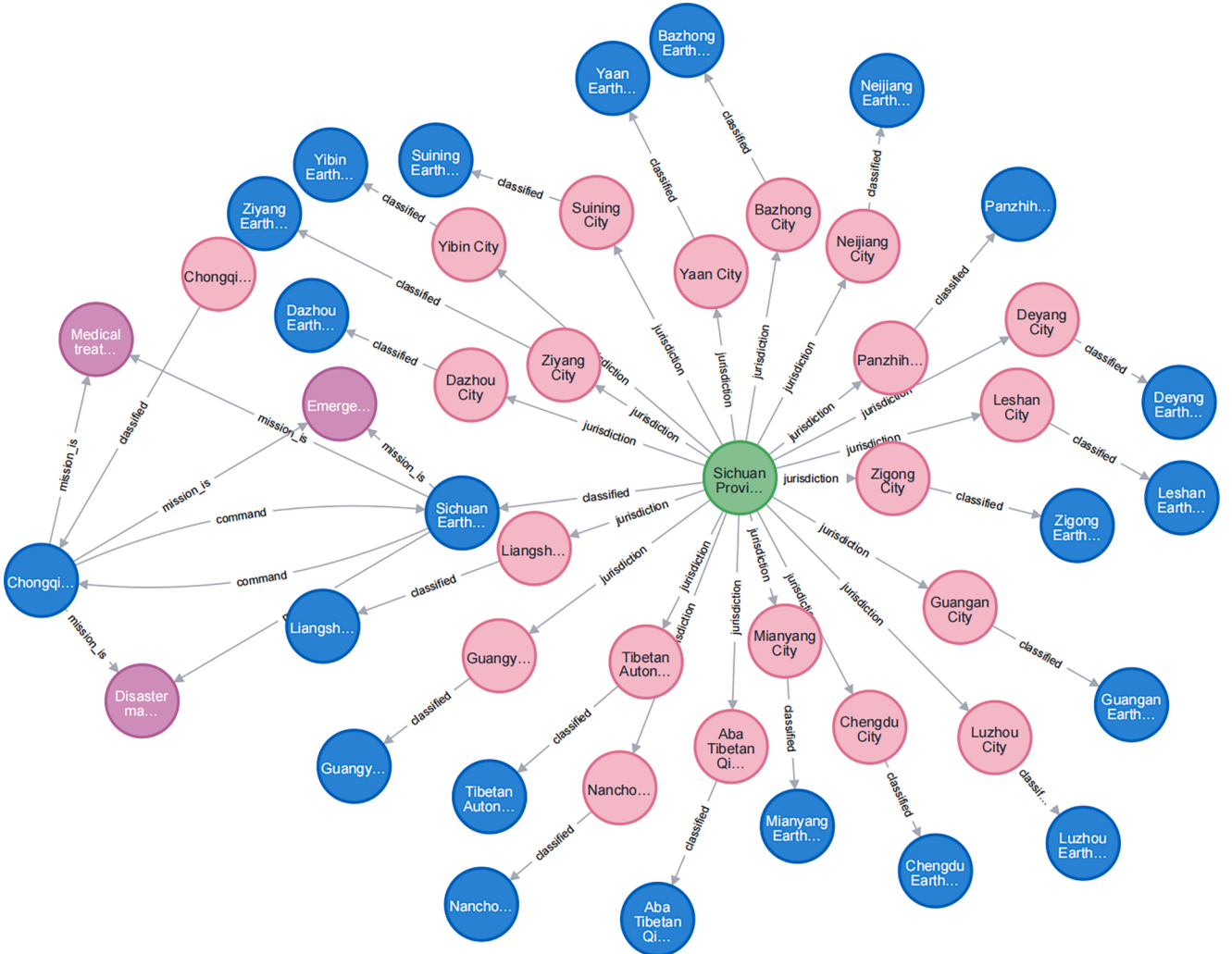


Fig 10. Query mechanism for emergency rescue knowledge in the Chengdu-Chongqing urban agglomeration.

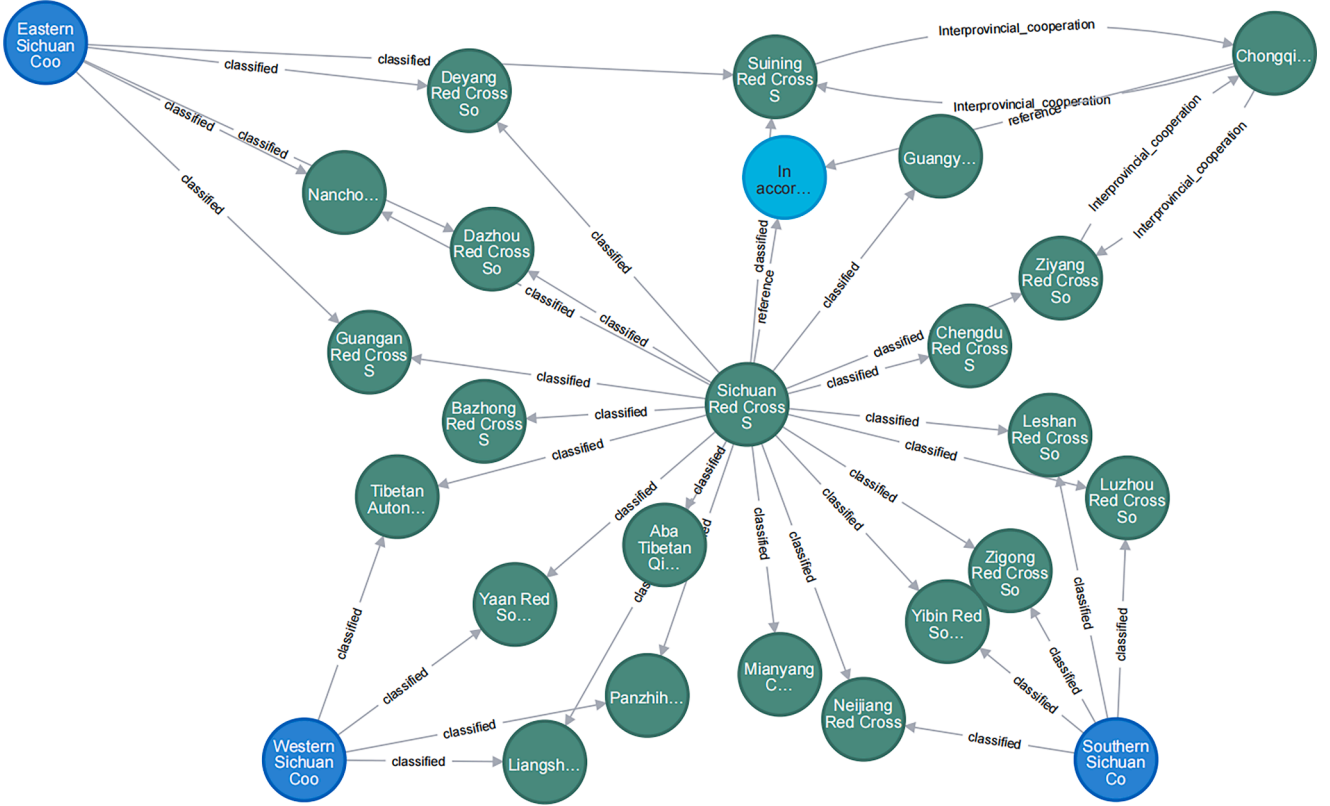


Fig 11. Schematic diagram of the Red Cross Society emergency rescue collaboration.

perform relatively better, while for Recall, the BERT and BERT-BiLSTM-CRF models perform relatively better. The final F1-value case obtained, the BERT-BiLSTM-CRF models also work relatively better than the other three models. This demonstrates that the BERT-BiLSTM-CRF models run entity recognition with more excellent and superior results.

3.2. Knowledge fusion

Due to the variability in entity names and categories across Earthquake Emergency Plans issued by different cities, which leads to the existence of multiple expressions of an entity, resulting in a redundant situation in the knowledge extraction process, it is necessary to perform knowledge fusion operations on the data to determine whether the entities under different expressions refer to the same underlying entity. Thus, the redundancy is eliminated, the ambiguity between entities is eliminated, and the quality of the data is ensured. For example, “end of emergency response” and “end of response” both represent the time description of the end of emergency rescue and transfer to the recovery and reconstruction phase. “Municipal Health Commission” and “Municipal Health Committee” refer to the same rescue entity. Therefore, performing knowledge fusion can make entity definitions more comprehensive and accurate, reduce the redundancy of data, and then construct a knowledge graph with higher accuracy.

The semantic similarity thresholds of A and B are calculated as shown in Eq. (10).

$$Sn(A, B) = \frac{A \bullet B}{|A| \times |B|} = \frac{\sum_{i=1}^n (A_i \times B_i)}{\sqrt{\sum_{i=1}^n (A_i)^2} \times \sqrt{\sum_{i=1}^n (B_i)^2}} \quad (10)$$

A and B are word frequency vectors. After being converted into a vector space, the similarity of the seismic event related entities in the vector space is evaluated using the cosine similarity. The higher cosine

similarity indicates a higher degree of similarity [38]. After experimental analysis, it is obtained that the knowledge fusion is better when the threshold value is 0.6, as shown in Table 5.

4. Knowledge graph visualization and retrieval

The Outline of the Construction Plan for the Chengdu-Chongqing Urban Agglomeration Twin city Economic Circle [39] emphasizes in Chapter 11, Section 4 that “the prevention and treatment of major disaster events and emergency rescue joint mechanism should be improved, a two-hour emergency rescue circle should be created, and the integration of disaster prevention, mitigation and relief should be promoted.” Therefore, this study focuses on three important aspects of the coordinated rescue of earthquake disaster emergency response in urban agglomeration: rescue and relief, scheduling of relief materials, and medical treatment.

Through the above knowledge extraction and knowledge fusion, this study transforms the data into structured “entity-relationship-entity” triad data to form the knowledge map of urban agglomeration earthquake disaster emergency response and coordination. Using the Neo4j graph database, the knowledge related to the cooperative rescue in earthquake disasters is visualized and structured, describing the rescue entities, policy documents, rescue actions, rescue information, rescue supplies, emergency response levels, and the interrelationships among the entities involved in coordinated rescue efforts in the aftermath of a disaster. The Neo4j graph database facilitates the visualization and display of the knowledge map from multiple dimensions of the concepts, improving the overall efficiency of graphical data modeling. This approach improves the overall efficiency and flexibility of graph data modeling, with the graph-based storage method offering superior query speeds compared to relational databases.

Cities within the Chengdu-Chongqing urban agglomeration are required to participate in coordinated emergency rescue operations in the event of a disaster by running the Neo4j graph database to query the



Fig 12. Knowledge graph for Ganzi earthquake disaster emergency rescue collaboration.

relevant nodes between the emergency rescue collaboration actions in Sichuan and Chongqing. There are 37 nodes (entities) and 36 edges (relationships) for this query situation. The cities appearing in Fig. 10 need to carry out rescue collaboration work after the earthquake disaster. The rescue collaboration work in the face of an earthquake disaster is done within the urban agglomeration to provide rescue supplies, rescue information, rescue entities, and medical treatment. In order to minimize the losses and impacts of the earthquake disaster on the inner cities.

The Red Cross Society also plays an important role at the level of emergency coordinated rescue and dispatch of rescue supplies in earthquake disasters. The five southwestern provinces and regions, including Sichuan Province and Chongqing City, signed the China Red Cross Society Emergency Relief Southwest Collaboration Zone Emergency Rescue Linkage Working Mechanism in 2023 and organized a comprehensive emergency response exercise for the Southwest Collaboration Zone, compiled the China Red Cross Society Emergency Relief Southwest Collaboration Zone Emergency Rescue Linkage Working Mechanism Working Program, the China Red Cross Society Southwest Operational Area Joint Rescue Emergency Response Plan, and establishment of improved mechanisms for joint command and scheduling, work exchanges, and coordination, carried out joint training exercises for rescue forces and other activities, basically constructed the Southwest Collaboration Zone emergency rescue work institutional mechanism framework, laying the foundation for the advancement of regional emergency rescue work [40].

Meanwhile, in 2024, Sichuan Province carried out regional

emergency cooperation exercises in the East and West of Sichuan as well as in the South of Sichuan, constructing the collaboration areas in these regions, and laying a solid foundation for rescue collaboration work within the urban agglomeration. The resources provided by the Red Cross Society, including relief family kits, food, and folding beds, can also help people in the affected areas promptly. Relief collaboration mechanisms operate within Sichuan Province and between Sichuan Province and Chongqing City. Fig. 11 depicts the relationships and linkages within Sichuan Province and the Red Cross Society in Chongqing City, which is divided into three major collaboration areas: Western Sichuan Collaboration Zone, Southern Sichuan Collaboration Zone, and Eastern Sichuan Collaboration Zone. A cross-provincial emergency response linkage mechanism also exists between Chongqing, Suining, and Ziyang. Following the earthquake disaster in Ganzi Tibetan Autonomous Prefecture, the Red Cross Society of Ganzi Tibetan Autonomous Prefecture and the Red Cross Societies and Blue Sky rescue teams of Ya'an, Panzhihua, and Liangzhou, which are also part of the West Sichuan Collaboration Zone, assisted in dispatching materials, providing medical aid, and mobilizing rescue teams for emergency response. This exemplifies the effective coordination between cities in emergency relief efforts.

5. Case study

In this study, the 2022 Luding earthquake event is input into the constructed knowledge graph for rescue collaboration in earthquake disaster emergency response to validate its effectiveness. Luding

```

labels: {
  "mission"
},
properties: {
  "name": "DISASTER REPORT:No.PRIOR DISPOSAL:Dispatch rescue teams in accordance with earthquake emergency response procedures.ORGANIZATION,COMMAND AND EMERGENCY RESPONSE:Develop
emergency rescue action plans, organize and coordinate emergency rescue forces, conduct personnel search and rescue and disaster relief operations, guide disaster-affected areas in
organizing self-rescue and mutual aid efforts, and oversee the mobilization, organization, deployment, and management of social emergency rescue forces and volunteer teams.
Coordinate with emergency rescue teams both within and outside the province, as well as within and outside the country.EMERGENCY RESPONSE:(1)Disaster Situation Collection and
Analysis. Utilize satellite and drone remote sensing, aerial reconnaissance, and ground surveys to obtain on-site imagery. Employ comprehensive monitoring methods and simulation
systems to rapidly assess the extent of damage in disaster-affected areas. Organize the collection and consolidation of disaster situation information from various regions and
industries. When necessary, publicly disclose disaster situation reporting methods through multiple channels such as radio, television, and the Internet to mobilize all sectors of
society to participate in disaster situation collection efforts.(2)Personnel Search and Rescue. Organize and guide local areas to conduct self-rescue and mutual aid. Promptly
deploy comprehensive fire rescue teams and work safety rescue teams. Coordinate with the Peoples Liberation Army, Armed Police Forces, reserve forces, and militia stationed in the
disaster-affected area and surrounding regions. Allocate large-scale machinery, demolition and support tools, and life detection devices, among other specialized rescue equipment,
to rush to the disaster-affected area and conduct household inspections and personnel search and rescue operations.(3)Medical treatment. Organize medical teams to rush to the
disaster area to rescue injured people. If necessary, establish temporary hospitals or medical stations to provide on-site treatment. Strengthen the organization and dispatch of
ambulances, medical equipment, medicines, and plasma. Increase the supply of medical equipment and medicines to severely affected areas. According to needs, transfer severely
injured patients for treatment in other locations. Properly handle the remains of the deceased, handle the aftermath for the deceased and comfort their families, and promptly
conduct psychological counseling and crisis intervention for the public.(4)Evacuation and relocation of affected residents. Organize the evacuation of affected and at-risk
residents, encourage residents to seek shelter with relatives or friends, promptly open emergency shelters, establish temporary and safe centralized relocation sites, and conduct
safety inspections and patrols at these sites. Set up centralized reception points (warehouses) for disaster relief supplies, coordinate the transportation of food, drinking water,
clothing, blankets, tents, folding beds, and portable toilets, and implement disaster relief assistance in accordance with relevant regulations.END OF EMERGENCY
RESPONSE:No.RESTORATION AND RECONSTRUCTION:Carry out post-disaster recovery and reconstruction work."
},
"elementId": "255"
}

```

(a)

```

MATCH (a:organization{name:'Medical Epidemiology Team of Ganzi'})-[b:Classified]→(c) RETURN c;

```

```

{
  "identity": 977,
  "labels": [
    "sector"
  ],
  "properties": {
    "name": "It is led by the State Health Commission, and consists of the State Development and Reform Commission, the State
Bureau of Economy and Informatization, the State Bureau of Agriculture, Animal Husbandry and Rural Affairs, the State Bureau
of Market Supervision, the State Bureau of Forestry and Grassland Management, the State Red Cross, and the peoples governments
of the affected counties (cities).",
  },
  "elementId": "977"
}

```

(b)

Fig 13. Emergency rescue collaboration retrieval framework for the Ganzi earthquake disaster.

County, located in the Ganzi Tibetan Autonomous Prefecture of Sichuan Province, was the area most seriously impacted by the earthquake, which had a magnitude of 6.8. According to the rules mentioned in Section 2.2.3 of this study, it is necessary to immediately activate the first level response to the earthquake disaster, and carry out the emergency rescue actions according to the corresponding task flow in the knowledge graph for the first level response.

Querying the nodes related to earthquake rescue in Ganzi Tibetan Autonomous Prefecture in the knowledge graph, there are 59 nodes and 60 edges (relations). Among them, the Western Sichuan Collaboration Zone, to which Ganzi Tibetan Autonomous Prefecture belongs, carries out coordinated rescue. Ya'an City, Panzhihua City, and Liangshan Yi Autonomous Prefecture dispatch rescue entities and provide supplies such as food, folding beds, and other rescue supplies, and the Red Cross Society provides rescue supplies such as disaster relief family boxes and other necessary supplies. After the earthquake disaster occurred in Ganzi Tibetan Autonomous Prefecture, the Municipal Earthquake Relief Command issued rescue orders to the rescue entities and assigned specific rescue actions. Earthquake disaster emergency relief operations are carried out in the order of disaster report – prior disposal – organization, command and emergency response – emergency response – end of emergency response – restoration and reconstruction. The Earthquake Agency, Emergency Management Department, and other relevant

agencies rapidly acquire natural disaster information and promptly report and assign it to the rescue entities for subsequent rescue actions. Fig. 12 highlights the specific rescue actions within Ganzi Tibetan Autonomous Prefecture, along with the corresponding rescue actions of the rescue entities, making it easier for the entities to access information about their respective actions.

Members and specific tasks within the corresponding organization can be queried using the knowledge graph constructed in this study. To query the knowledge in Neo4j, take the Ganzi earthquake as an example. Enter the code:

(MATCH(a:organization {name:'Emergency and Disaster Relief Team of Ganzi '})- [b:Missionis] -> (c) RETURN c;)

Here, a represents the rescue entity named “Emergency and Disaster Relief Team of Ganzi”, b represents the relationship “Missionis”, and the Cypher query outputs c, which corresponds to the specific disaster relief tasks assigned to the “Emergency and Disaster Relief Team of Ganzi” after the earthquake. Additionally, this study categorises tasks in chronological order, which helps rescue entities clarify rescue actions. And we can get the specific mission of “Emergency and Disaster Relief Team of Ganzi”. The query schematic is shown in Fig. 13(a). Similarly, the query for the members of “Ganzi Medical Epidemiology Team” is shown in Fig. 13(b).

Through the verification of specific cases, it can be found that the

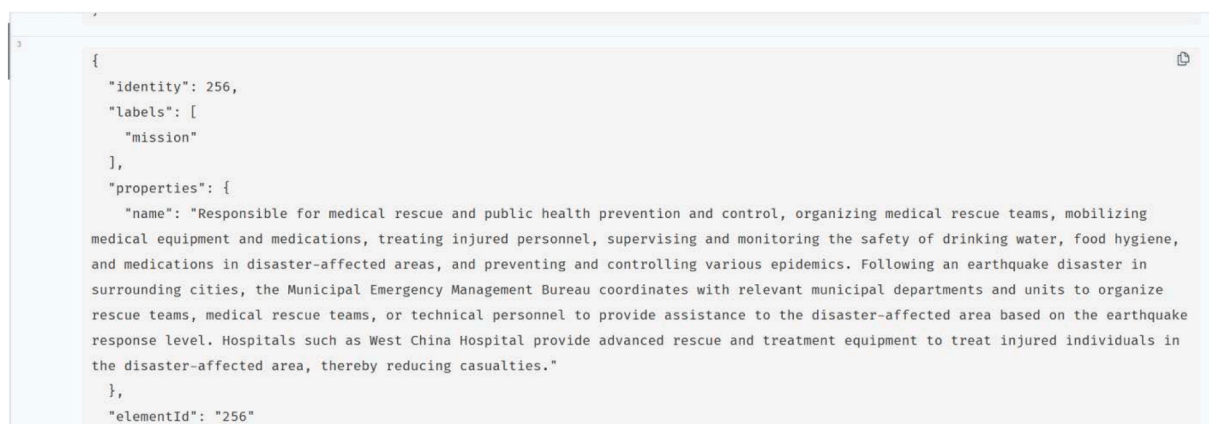


Fig 14. Inquiry status of the Chengdu medical epidemic prevention team.

knowledge mapping for cooperative rescue in earthquake disaster emergency response within urban agglomeration, constructed in this study can clarify the actions and responsibilities of rescue entities, strengthen the intercity cooperative rescue situation within the urban agglomeration, and provide decision-making support for the rescue actions in earthquake disasters, which can effectively help the cooperative rescue work in earthquake disasters.

Since most of the data used to construct the knowledge graph comes from official government websites in this study, there is applicability of the concepts and attributes of its ontology. Together with the knowledge fusion, the knowledge graph constructed in this study can be utilized across multiple domains. Not only can it be queried for earthquake disasters, but it can also be queried for other types of natural disasters. Furthermore, although the construction of this knowledge graph is targeted at the Chengdu-Chongqing urban agglomeration, it is extendable to other urban agglomerations, thereby enhancing its applicability.

Taking Chengdu as an example, a query of the rescue actions of the Chengdu Medical Epidemic Prevention Team in Neo4j is shown in Fig. 14. In the event of an earthquake disaster in surrounding provinces and cities, timely responses will be made based on the emergency response level. Major hospitals in Chengdu, such as West China Hospital, will promptly provide rescue medicines, advanced medical supplies, and equipment to the disaster area, and dispatch rescue teams to treat injured people in the disaster area to reduce casualties. It can be demonstrated that the knowledge graph constructed in this study is not only applicable to the Chengdu-Chongqing urban agglomeration but also holds applicability to other urban agglomerations to a certain extent.

6. Conclusions

From the lack of experience in emergency rescue collaboration during the 2008 Wenchuan earthquake in China, to the mature urban agglomeration emergency rescue collaboration mode demonstrated by the 2025 Shigatse earthquake, significant reductions in casualties and economic losses have been achieved. This fully confirms the importance and necessity of urban agglomeration emergency rescue collaboration in earthquake disasters. Knowledge graphs offer an innovative approach for acquiring, storing, and visualizing earthquake disaster information, as well as a scientific basis for decision-making in emergency rescue collaboration. This study explores the methodology and process of constructing an urban agglomeration earthquake disaster emergency rescue collaboration knowledge graph. Drawing from Earthquake Disaster Emergency Plans released by various cities within the Chengdu-Chongqing urban agglomeration in China, emergency rescue data from the Sichuan and Chongqing Red Cross Societies, and historical disaster rescue information from the Blue Sky rescue team, six types of entities

including rescue entities, policy documents, rescue materials, rescue actions, emergency response levels, and rescue information are extracted in this study. These entities are subsequently classified and defined in terms of concepts, relationships, rules, and instances. Through knowledge extraction and knowledge fusion, the data layer of the knowledge graph is constructed. The 2022 Luding earthquake disaster in China is selected as a typical case, and the constructed knowledge graph is validated by inputting the case data. The results demonstrate that the knowledge graph effectively illustrates the emergency response mechanism and rescue overview of cross-city emergency rescue collaboration within the urban agglomeration in the earthquake disaster, demonstrating high reliability and effectiveness of the knowledge graph that was constructed in this study. It also exhibits broad applicability in terms of concepts, relationships, rules, and instances, effectively describing entity concepts and revealing inter-entity relationships. The construction and application of this knowledge graph can provide intelligent decision-making support for urban agglomerations in responding to earthquake disasters, reducing the time required for decision-makers to formulate emergency strategies and enhancing the performance of government disaster rescue efforts. Furthermore, it facilitates the development of urban agglomeration emergency management systems and offers new ideas and methods for earthquake disaster rescue and mitigation.

Ethics Statement

Not applicable because this work does not involve the use of animal or human subjects.

CRediT authorship contribution statement

Lin Zhang: Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Qichen Wang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **YanJun Guo:** Writing – review & editing, Supervision, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Xiangliang Tian:** Writing – review & editing. **Lin Qi:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] W. Lyu, J.Q. Tang, P.J. Zhao, et al., Review and prospect of research on Human mobility under Earthquake disasters in China, *J. Catastrophol.* (2024) 1–9.
- [2] Y. Chen, L.Y.Z. Zhang, W.B. Song, Improving emergency preparedness with a live collaboration exercise model for first responders, *Sci. Technol. Prog. Policy.* 39 (09) (2022) 33–41.
- [3] K. Kang, T. Chen, H.Y. Yuan, Cooperative scheduling model for multilevel emergency response teams, *J. Tsinghua Univ.: Sci. Technol.* 56 (8) (2016) 830–835. +843.
- [4] D. Sjöberg, M. Inzunza, Improving emergency preparedness with a live collaboration exercise model for first responders, *Int. J. Emerg. Manag.* 17 (3) (2022) 195–216.
- [5] B. Barth, G.C. Kabbinahithilu, J. Marchal, et al., Web-based solutions for communication and knowledge management in disaster situations, *IEEE Internet Comput.* 27 (1) (2023) 53–59.
- [6] L. Zhu, X.Y. Duan, L.Y. Bai, SSQTKG: a subgraph-based semantic query approach for temporal knowledge graph, *Data Knowl. Eng.* (2024) 102372.
- [7] Y.C. Zhang, X.J. Kong, Z.H. Shen, et al., A survey on temporal knowledge graph embedding: models and applications, *Knowl. Based Syst.* 304 (2024) 112454.
- [8] L. Wang, C.H. Sun, C.Y. Zhang, et al., Application of knowledge graph in software engineering field: a systematic literature review, *Inf. Softw. Technol.* 164 (2023) 107327.
- [9] M.Z. Chen, Z.X. Tao, W.T. Tang, et al., Enhancing emergency decision-making with knowledge graphs and large language models, *Int. J. Disaster Risk Reduct.* (2024) 104804.
- [10] C.G. Chen, J.Q. Hu, Z.C. Han, et al., Knowledge graph based early warning method for accident evolution in natural gas pipeline station abroad for harsh environmental conditions, *J. Tsinghua Univ.: Sci. Technol.* 62 (06) (2022) 1081–1087.
- [11] Y.Y. Chen, L.B. Zhang, J.Q. Hu, et al., An emergency task recommendation model of long-distance oil and gas pipeline based on knowledge graph convolution network, *Process Saf. Environ. Prot.* 167 (2022) 651–661.
- [12] Z. Yan, X. Zhou, Q.Y. Zhang, et al., Predicting multi-subsequent events and actors in public health emergencies: an event-based knowledge graph approach, *Comput. Ind. Eng.* 187 (2024) 109852.
- [13] M.K. Li, C. Yuan, K.J. Li, et al., Knowledge management model for Urban Flood emergency response based on multimodal Knowledge graphs, *Water* 16 (2024) 12.
- [14] F. Simone, S.M. Ansaldi, P. Agnello, et al., Knowledge in graphs: investigating the completeness of industrial near miss reports, *Saf. Sci.* 168 (2023) 106305.
- [15] N. Wang, X. Yang, J.H. Chen, et al., Hazards correlation analysis of railway accidents: a real-world case study based on the decade-long UK railway accident data, *Saf. Sci.* 166 (2023) 106238.
- [16] Y.R. Yang, Z.D. Cao, P.F. Zhao, et al., Constructing public health evidence knowledge graph for decision-making support from COVID-19 literature of modelling study, *J. Saf. Sci. Resil.* 3 (2) (2021) 146–156.
- [17] Z.X. Tao, X.H. Liu, Y. Li, et al., Intelligent emergency assisted decision-making method based on standard digitalization: hazardous chemical accidents in industrial parks, *J. Saf. Sci. Resil.* 1 (6) (2025) 79–92.
- [18] L. Chen, M. Chang, H.N. Yang, et al., Comprehensive evaluation and optimal management of extreme disaster risk in Chinese urban agglomerations by integrating resilience risk elements and set pair analysis, *Int. J. Disaster Risk Reduct.* 111 (2024) 104671.
- [19] X.Y. Xiao, S.Y. Mei, R.Q. Liu, et al., Impact of the COVID-19 pandemic on the development of core cities in the Beijing-Tianjin-Hebei urban agglomeration—taken Beijing, Tianjin, and Shijiazhuang as examples, *J. Tsinghua Univ.: Sci. Technol.* 63 (06) (2023) 994–1002.
- [20] W.Y. Du, C. Liu, Q.Y. Xia, et al., OFPO & KGFP: ontology and knowledge graph for flood process observation, *Environ. Model. Softw.* 185 (2025) 106317.
- [21] S.Z. Phua, M. Hofmeister, Y.-K. Tsai, et al., Fostering urban resilience and accessibility in cities: a dynamic knowledge graph approach, *Sustain. Cities Soc.* 113 (2024) 105708.
- [22] K. Janowicz, P. Hitzler, W. Li, et al., Know, Know where, KnowWhereGraph: a densely connected, cross-domain knowledge graph and geo-enrichment service stack for applications in environmental intelligence, *AI Mag.* 43 (2) (2022) 270.
- [23] N. Xiao, B. Peng, X. Li, et al., Research on the construction and implementation of power grid fault handling knowledge graphs, *Energy Rep.* 2 (9) (2023) 182–189.
- [24] J.J. Wang, W. Zhu, X.P. Yin, et al., Research on knowledge graph construction method for emergency response of hazardous chemicals accident, *J. Saf. Sci. Technol.* 20 (03) (2024) 46–52.
- [25] B.H. Yang, C.L. Zhou, S.J. Li, et al., A Chinese named entity recognition method for landslide geological disasters based on deep learning, *Eng. Appl. Artif. Intell.* 139 (2025) 109537.
- [26] W. Chen, H.B. Zhang, L.K. Comfort, et al., Exploring complex adaptive networks in the aftermath of the 2008 Wenchuan earthquake in China, *Saf. Sci.* (2020) 125.
- [27] Z.H. Wang, M. Huang, C.X. Li, et al., Intelligent recognition of key earthquake emergency Chinese information based on the optimized BERT-BiLSTM-CRF algorithm, *Appl. Sci.-Basel* 5 (13) (2023) 3024.
- [28] P.J. Shi, H. Zhang, China's response to the catastrophe: the experience of Wenchuan earthquake, *J. Tsinghua Univ.: Philos. Soc. Sci. Ed.* 28 (03) (2013), 96–113+160.
- [29] D.G. Liu, L. Cheng, MAKG: a maritime accident knowledge graph for intelligent accident analysis and management, *Ocean Eng.* 312 (3) (2024) 119280.
- [30] D. Varshney, A. Zafar, N.K. Behera, et al., Knowledge graph assisted end-to-end medical dialog generation, *Artif. Intell. Med.* 139 (2023) 102535.
- [31] Y.F. Liu, S.Q. Wei, H.J. Huang, et al., Naming entity recognition of citrus pests and diseases based on the BERT-BiLSTM-CRF model, *Expert Syst. Appl.* 234 (2023) 121103.
- [32] X.Y. Zhang, G.Y. Liu, Y. Xu, et al., A knowledge graph-based inspection items recommendation method for port state control inspection of LNG carriers, *Ocean Eng.* 1 (33) (2024) 119434.
- [33] W. Li, Y.J. Du, X.Y. Li, et al., UD BBC: named entity recognition in social network combined BERT-BiLSTM-CRF with active learning, *Eng. Appl. Artif. Intell.* 116 (2022) 105460.
- [34] J.R. Chen, Y.Q. Lu, Y. Zhang, et al., A management knowledge graph approach for critical infrastructure protection: ontology design, information extraction and relation prediction, *Int. J. Crit. Infrastruct. Prot.* 43 (2023) 100634.
- [35] F.G. Zhou, C. Wang, J.P. Wang, Named entity recognition of ancient poems based on Albert-BiLSTM-MHA-CRF model, *Wirel. Commun. Mob. Comput.* (2022) 6507719.
- [36] C.K. Xia, S.B. Li, Chinese named entity recognition method based on BERT-BiLSTM-MHA-CRF model, *Comput. Digit. Eng.* 51 (09) (2023) 2087–2091. +2102.
- [37] Z. Yan, X. Zhou, Q.Y. Zhang, et al., Predicting multi-subsequent events and actors in public health emergencies: an event-based knowledge graph approach, *Comput. Ind. Eng.* 187 (2024) 109852.
- [38] P.Y. Qiu, L.K. Pang, Y. Luo, et al., Earthquake event knowledge graph construction and reasoning, *Geomat. Nat. Hazards Risk.* 15 (1) (2024).
- [39] China Government Network. The Central Committee of the Communist Party of China and the State Council issued the Outline of the Plan for the construction of the Chengdu-Chongqing Urban agglomeration twin-city Economic Circle [EB/OL], 2021. <https://www.gov.cn/>.
- [40] The 2023 Work Conference of the Southwest Regional Collaboration Mechanism for Red Cross Emergency Rescue was held in Luzhou, Sichuan, China., 2024. EB/OL, <https://www.redcross.org.cn/>.